POLITECNICO DI TORINO

Master's Degree in Computer Engineering

Artificial Intelligence and Data Analytics track

From pixels to financial portfolios: generating realistic market scenarios with variational autoencoders

Thesis summary

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Thesis carried out in collaboration with Intesa Sanpaolo — Academic Year 2025–2026

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1. Context and research problem

The correlation matrix of asset returns underpins much of quantitative portfolio analysis, from Markowitz onwards. Estimating it from historical series alone, however, involves two distinct and structural limitations.

The first is *statistical noise*. When the number of assets N is of the same order of magnitude as the length T of the series used for estimation, the empirical matrix becomes unstable and its spectrum fills with spurious eigenvalues, a phenomenon systematically studied by Random Matrix Theory.

The second, and deeper, is *sampling limitation*: the past offers a single trajectory among all those that would have been possible. Market configurations never observed in the specific available sample, yet plausible in statistical terms, remain inaccessible to any purely historical analysis.

Hence the research question of this thesis: is it possible to build a deep generative model that, having learnt how correlations organise themselves in real data, produces new synthetic correlation matrices — mathematically valid and structurally indistinguishable from real ones — with which to populate a scenario space richer than the one history alone can provide? The closest precedent in the literature is CorrGAN (Marti, 2020), which addresses the same problem with adversarial networks; the work of Caprioli et al. (2024) instead uses a Variational Autoencoder on historical matrices with resampling in the latent space, an approach methodologically akin to the one adopted here.

2. Data and pre-processing

The empirical foundation is a dataset of adjusted closing prices of the S&P 500 index over the twenty-year period 2000–2020. To guarantee continuity of the series and avoid sampling distortions, the universe was filtered by requiring uninterrupted quotations over the entire horizon: $N = 362$ stocks survive. The GICS sector classification, required by the hierarchical structure analyses, was not available from a structured provider and was reconstructed through automatic classification by a large language model (360 out of 362 stocks classified).

Daily log-returns are computed from the prices and, from these, Pearson correlation matrices are estimated with a rolling window procedure. The window width is set to $T_w = 724$ days, exactly twice the number of assets: this choice fixes the spectrum ratio $q = N/T_w = 0.5$ by construction, making the noise level homogeneous over time and analytically governable through the Marchenko–Pastur bounds, and it guarantees that every matrix is full rank and therefore positive definite. With a stride of 10 days, 461 matrices are obtained.

A structural validity filter based on the empirical Perron–Frobenius property then acts on this raw set, requiring the first eigenvector (the *market mode*) to have components all of the same sign. The first 49 windows fail it; detailed analysis shows that these are all and only the windows containing observations preceding mid-October 2001, in which the only gold-mining stock in the universe (NEM) is systematically anticorrelated with the rest of the market in its typical safe-haven role. This is not a measurement artefact but the signature of an exceptional regime, and excluding it explicitly delimits the model's domain of validity: **412 valid matrices** remain, then randomly shuffled and split 288 / 82 / 42 across training, validation and test sets. Random shuffling, in place of chronological partitioning, neutralises the macroeconomic regime *distribution shift* which, in preliminary tests, prevented the deep architectures from generalising.

The positive semidefiniteness constraint and the Cholesky parametrisation

A decisive methodological step concerns the form of the input. A correlation matrix must be positive semidefinite (PSD): this is a global constraint on the spectrum, and cannot be imposed by acting on individual coefficients separately. In the first version of the pipeline the models directly reconstructed the lower triangle of the correlation matrix; spectral checks revealed the appearance of negative eigenvalues, and a single negative eigenvalue — however negligible in magnitude — is enough to make the Cholesky decomposition fail, and with it the entire scenario simulation.

The final pipeline solves the problem at its root by changing the object being learnt: each matrix is decomposed as $\mathbf{C} = \mathbf{L}\mathbf{L}^T$ and what is flattened and fed to the models is the lower triangular factor $\mathbf{L}$, not the matrix. Whatever vector the decoder produces, the matrix $\hat{\mathbf{L}}\hat{\mathbf{L}}^T$ recomposed from it is a Gram matrix and therefore PSD by construction, independently of the quality of learning; a final renormalisation to unit diagonal brings the result back into the correlation domain without altering the signs of the eigenvalues. The canonical input is therefore a vector of $D = N(N+1)/2 = 65\,703$ elements.

3. Architectures and experimental protocol

The comparison follows an incremental path across four models, all trained on the same Cholesky parametrisation and evaluated at equal latent dimension K:

1. **Principal Component Analysis**, a linear benchmark that sets the upper limit of linear compression;
2. **Linear Autoencoder**, a single dense layer with neither bias nor activations, built to replicate that subspace and empirically verify the Bourlard and Kamp theorem;
3. **Deep Autoencoder**, a funnel design $2048 \to 1024 \to 512 \to K$ with LeakyReLU activations, introducing non-linearity;
4. **Variational Autoencoder**, which shares the funnel encoder of the previous model but replaces the deterministic bottleneck with a probabilistic inference module ($\boldsymbol{\mu}$, $\log \boldsymbol{\sigma}^2$, reparameterization trick).

The VAE reconstruction loss is not the usual mean squared error but the **Jeffreys Divergence** between the Gaussian distributions associated with the original and the reconstructed matrix. The motivation is precise: MSE treats coefficients as independent quantities and ends up dominated by the larger correlations, effectively blind to the smaller eigenvalues of the spectrum — precisely those that identify minimum-variance portfolios. The inverse matrices in the Jeffreys formulation reverse this hierarchy and weigh the error in relative terms along every direction of the spectrum. Because the reconstruction term is itself a divergence, it is homogeneous with the Kullback–Leibler regularisation term and the weight β remains fixed at 1: the formulation used is therefore the canonical ELBO.

Evaluation combines two families of metrics: spatial errors (MSE, MAE, Frobenius norm) and topological metrics built on the *Minimum Spanning Tree* (percentage of shared edges, Jaccard index, overlap of the nodes with highest betweenness centrality, differences in average path length). The latter verify that a reconstruction accurate "on average" has not nonetheless altered the hierarchy of links between stocks.

4. Results on historical reconstruction

The measurements empirically confirm the Bourlard and Kamp theorem: PCA and the Linear Autoencoder converge, at equal K, on the same error limit. The Deep Autoencoder clearly surpasses that threshold at every K and shows no sign of overfitting: at $K = 20$ its test MSE (7.19×10^{-5}) is almost four times lower than PCA's (2.73×10^{-4}). The advantage of non-linearity emerges as one of *representational efficiency*: the AE already reaches with $K = 10$ the same quality that PCA attains only with $K = 100$.

The Variational Autoencoder sits in an intermediate position. At $K = 20$ it achieves an MSE of 1.55×10^{-4}, **lower** than both PCA and the Linear Autoencoder, while remaining higher than the deterministic Deep Autoencoder. MAE and Frobenius norm return exactly the same ranking, which indicates that the ordering is not an artefact of MSE's sensitivity to outliers. The topological picture is even sharper and consistent: on MST edge overlap the VAE recovers two thirds of the distance between the linear models and the Deep Autoencoder (88.4% against PCA's 83.0% and the AE's 91.1%). The cost that variational regularisation imposes on reconstruction fidelity is therefore contained, and smaller than a first intuition would suggest.

Three runs at $K \in \{10, 20, 30\}$, identical in every other hyperparameter, reveal a peculiar behaviour of the VAE: its reconstruction error *worsens* slightly as K grows, the opposite of the other three models. Analysis of the latent space activity spectrum suggests an explanation: in all three configurations only about six dimensions are effectively active, while the remaining ones undergo *posterior collapse*. The effective dimensionality of the correlation manifold therefore appears not to be a hyperparameter to be tuned but rather a property of the data that the model discovers on its own; adding nominal dimensions does not increase the capacity actually used and merely widens the space over which conformity to the prior must be satisfied. The choice of $K = 20$ for the final model — the elbow of the reconstruction curve — is thus robust.

5. Stochastic scenario generation and validation

It is on the generative side that the VAE shows its distinctive value. The deterministic Deep Autoencoder, however superior in pure reconstruction, has a fragmented latent space devoid of meaning outside the observed points: sampling a random vector from it produces output with no economic or financial coherence. Kullback–Leibler regularisation, combined with sampling noise, instead shapes a continuous and dense latent space in which meaningful sampling is possible. The most effective synthesis of the results is this complementarity: *the AE to remember, the VAE to imagine.*

The generation procedure actually adopted, however, is not a simple sampling from the prior $\mathcal{N}(\mathbf{0}, \mathbf{I})$, which would produce some arbitrary plausible scenario, disconnected from the present. The objective is more specific: to build a distribution of realistic scenarios of the correlation structure *at a one-month horizon starting from the market state observed today*, to be assessed from a risk perspective. The technique used is a **historical block bootstrap in the latent space**, articulated in four steps: the encoder reconstructs the historical latent trajectory day by day; its daily increments are computed; a block of 21 consecutive increments (one working month) is randomly sampled and its cumulative jump computed; the jump is added to the current state and decoded, yielding a PSD matrix by construction. Repeating this a thousand times yields a distribution of possible evolutions, anchored to the present yet diversified by block-sampled historical variability.

The one thousand scenarios so produced were validated against the five stylized facts of financial correlation matrices, each translated into a scalar indicator comparable with today's matrix and with the range observed over the 412 real matrices:

1. **Pairwise distribution skewed towards positive values**: mean between 0.4609 and 0.5352 across all scenarios, against 0.4512 for today's matrix.
2. **Spectrum and Marchenko–Pastur**: market eigenvalue λ_1 in the range $[172.4,\ 197.2]$ against today's 169.1, roughly two orders of magnitude beyond the upper edge of the bulk, accompanied by 7–9 sector modes.
3. **Perron–Frobenius property**: the minimum component of the market mode is always positive, with a larger margin from the violation threshold than today's matrix itself.
4. **Hierarchical sector structure**: the ratio between average intra-sector and inter-sector correlation is always greater than 1 and within the historical range; the dendrograms visually reproduce the same sector colour blocks as the real matrix.
5. **Scale-free MST topology**: the power-law exponent lies between 2.053 and 2.306, inside the range $2 < \gamma < 3$ expected from the literature.

The outcome is that **all 1000 scenarios satisfy the five criteria simultaneously**, and all matrices are positive definite, with positive semidefiniteness guaranteed by construction by the Cholesky parametrisation. Detailed graphical inspection of a single scenario corroborates the picture on those objects no scalar indicator can summarise: cosine similarity with respect to today's matrix is 0.9999 on the market mode and 0.9893 and 0.9883 on the two subsequent sector modes.

6. Limitations and future work

The work explicitly declares its methodological limitations. The aggregate verification reduces each stylized fact to a scalar indicator, while full visual inspection remains limited to a single representative scenario; the one thousand scenarios moreover derive from heavily overlapping 21-day blocks, with an effective sample size below one thousand, and the entire verification is anchored to a single starting market state. Random shuffling, while addressing distribution shift, does not remove the informational overlap between adjacent windows (up to 98.6% of shared observations), which represents a limited but real risk of optimism in the reported generalisation metrics. Finally, the model comparison does not cover all latent dimensions uniformly for the VAE, and the GICS sector classification was not validated against official sources.

The natural continuation is to apply the generated scenario distribution to portfolio risk management, comparing the resulting picture with the one obtainable from historical data alone. Next come the strengthening of statistical validation — repeating the verification from a plurality of initial states distributed across the twenty-year horizon and extending it from scalar indicators to formal distributional goodness-of-fit tests — a direct comparison with alternative generative paradigms (CorrGAN, and more generally diffusion models), and the extension of the pipeline to other asset classes and to international equity universes.

7. Conclusion

The thesis has designed, implemented and empirically evaluated a complete framework for the compression, reconstruction and stochastic generation of realistic financial correlation matrices. The most significant result is that the Cholesky parametrisation structurally resolves the positive semidefiniteness constraint, and that on this basis the Variational Autoencoder pays a contained cost in reconstruction fidelity — indeed remaining superior to the linear benchmarks — in exchange for the property that makes stochastic generation possible. The simultaneous conformity of the one thousand generated scenarios to the five stylized facts supports the proposed architecture as a self-contained tool for producing mathematically coherent synthetic financial ecosystems.